Solution Requirements

Date	30 June 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register and log in using email and password before accessing personalized learning features.	High
2	Authorization levels	Students can access learning features, while administrators manage users, content, and system settings.	High

3	External interfaces	The system integrates with Google Gemini API, FastAPI backend, and a database to generate AI-powered educational responses.	High
4	Transactions processing	User queries are sent to the AI model, processed securely, and responses are stored in the user's learning history.	High
5	Reporting	The system provides quiz results, learning progress, summaries, and recommendation reports through the user dashboard.	Medium
6	Business rules, etc.	Every user query is validated before processing. Personalized recommendations are generated based on previous learning history.	High
7	Compliance to laws or regulations	The application protects user data, follows privacy policies, and securely stores user information.	Medium
8	*Other*	The system supports AI-based question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate AI responses quickly with minimal delay.	Response within **3–5 seconds**
2	Scalability	The system should support multiple users simultaneously without affecting performance.	Supports **500+ concurrent users**
3	Security & Data Privacy	User credentials and learning history should be securely stored and transmitted.	HTTPS communication and encrypted password storage
4	Reliability & Availability	The application should remain available with minimal downtime.	99% system availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy for students of all levels to use.	Responsive design with intuitive navigation
6	*Other*	The application should be maintainable, modular, and easy to deploy on different operating systems.	Compatible with Windows, Linux, and macOS